The Journal of the Acoustical Society of America

NOVEMBER 11 2026

Acoustic detection probability of harbour porpoises depends on behaviour, detection unit, and click-train classification

Dinah Hartmann ; Jamie D. J. Macaulay ; Laia Rojano-Doñate ; Jonas Teilmann ; Siri L. Elmegaard ; Peter T. Madsen ; Ursula Siebert ; Peter Tyack ; Len Thomas

Check for updates

J. Acoust. Soc. Am. 160, 4122–4136 (2026)
https://doi.org/10.1121/10.0047478

View Online
Export Citation

Articles You May Be Interested In

The whale binaural rings experiment: Long-range acoustic monitoring of North Atlantic right whales in the Gulf of St. Lawrence

J. Acoust. Soc. Am. (October 2026)

The use of seal scarers as a protective mitigation measure can induce hearing impairment in harbour porpoises

J. Acoust. Soc. Am. (December 2019)

Porpoise click classifier (PorCC): A high-accuracy classifier to study harbour porpoises (*Phocoena phocoena*) in the wild

J. Acoust. Soc. Am. (June 2019)

11 November 2026 21:14:56

ASA
LEARN MORE
Advance your science and career as a member of the
Acoustical Society of America

ASA
ACOUSTICAL SOCIETY OF AMERICA

JASA ARTICLE

Acoustic detection probability of harbour porpoises depends on behaviour, detection unit, and click-train classification

Dinah Hartmann,[1,2,a] Jamie D. J. Macaulay,[2] Laia Rojano-Doñate,[3,4] Jonas Teilmann,[3] Siri L. Elmegaard,[3] Peter T. Madsen,[4] Ursula Siebert,[3,5] Peter Tyack,[2] and Len Thomas[1]

[1]*Centre for Research into Ecological and Environmental Modelling, School of Mathematics and Statistics, University of St Andrews, St Andrews, KY16 8LZ, Scotland*

[2]*Scottish Oceans Institute, School of Biology, University of St Andrews, St Andrews, KY16 8LB, Scotland*

[3]*Marine Mammal Research, Department of Ecoscience, Aarhus University, Roskilde 4000, Denmark*

[4]*Center for Active Sensing with Sound, Department of Biology, Aarhus University, Aarhus 8000, Denmark*

[5]*Institute for Terrestrial and Aquatic Wildlife Research, University of Veterinary Medicine Hannover, Buesum 25761, Germany*

ABSTRACT:

Harbour porpoise (*Phocoena phocoena*) densities are estimated routinely from acoustic detections. Previous acoustic studies show the probability of detecting porpoises (or conversely false negative rate) depends strongly on porpoise behaviour, changing density estimates several-fold if unaccounted for. We quantify this in widely used automated click detectors (e.g., C-PODs), where post-processing is used to identify sequences of consecutive clicks ("click trains") and summarise results as detection-positive time units (e.g., click-positive seconds or minutes). We explored how post-processing choices change detectability using 228 h of fine-scale acoustic and movement data from tags on ten harbour porpoises in Danish waters. We simulated the click detection process, given assumed porpoise beam pattern and sound propagation, ran a custom click-train classifier on the resulting data and summarized results over different time units. Using porpoise-positive minutes after click-train classification as a metric, the most detectable behaviour (pelagic feeding) averaged 2.36 times [95% confidence interval 1.81–3.87] as detectable as the least (non-feeding) due to high click and turning rates. Click-train classification increased detectability differences between pelagic feeding and other behaviours. PAM studies may appreciably over- or under-estimate animal density; click-train classification may magnify this effect. We provide guidance on how these issues could be quantified and mitigated.

© *2026 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).* https://doi.org/10.1121/10.0047478

(Received 19 June 2026; revised 15 October 2026; accepted 19 October 2026; published online 11 November 2026)

[Editor: Kaitlin E. Frasier] Pages: 4122–4136

11 November 2026 21:14:56

I. INTRODUCTION

Harbour porpoises (*Phocoena phocoena*) produce stereotyped narrow-band high-frequency (130 kHz) echolocation clicks (Møhl and Andersen, 1973) for foraging, communication and navigation (Wisniewska *et al.*, 2016; Sørensen *et al.*, 2018). This makes passive acoustic monitoring (PAM) an attractive option for population assessment. However, echolocation rates vary substantially depending on behaviour, time of day, environmental conditions, and social interactions (Carlström, 2006; DeRuiter *et al.*, 2009; Sørensen *et al.*, 2018). The validity of PAM-based estimates of animal abundance hinges on an assumed relationship between animal abundance and number of detections. The probability of detecting animals or vocalisations corrects these estimates for an assumed proportion of missed detections within the monitoring area.

There are various frameworks for estimating animal density and abundance from fixed detector point transect PAM data (Marques *et al.*, 2013; Stevenson *et al.*, 2015). Where individual vocalisations ("cues"), such as clicks in the case of porpoises, are the unit of detection, density can be estimated using the equation

$$D = \frac{n_c(1-f_c)}{a_c P T r} \quad (1)$$

(Marques *et al.*, 2013), where n_c is the number of detected cues, f_c is the false positive rate, a_c is the monitored area size (defined to be large enough that no cues are detected outside it), P is the probability of detecting a cue in the monitored area, T is the time monitored and r is a cue rate (average number of cues produced per animal per unit time) to convert cue density into animal density. Cue rate is typically estimated from acoustic tag data (Warren *et al.*, 2017).

Detections from acoustic analysis are commonly summarised in time-based detection units such as "Detection- (or Click-) Positive Seconds/min/h/Days" in various contexts including density and abundance estimation [e.g., Amundin *et al.* (2022)], or studies of behaviour and animal

[a]Email: djh38@st-andrews.ac.uk

 ©Author(s) 2026.

https://doi.org/10.1121/10.0047478

disturbance by, e.g., noise from pile driving [e.g., Scheidat *et al.* (2011), Voß *et al.* (2023), and Calonge *et al.* (2024)]. Instead of counting cues (in our case single porpoise clicks), a deployment period is divided into time units (e.g., a number of minutes monitored), and each time unit is either counted as detection positive if it contains at least one cue detection or as detection negative if it contains no detections (Calonge *et al.*, 2024). This method assumes that each detection positive time unit contains a single group detection and negligible animal movement within a detection time unit (Kyhn *et al.*, 2012).

When using summary detection units, such as detection positive seconds, a similar framework as Eq. (1) can be used, with the exception that instead of a cue rate, detection positive time units need to be converted into animal density. This can be achieved using a mean group size and assuming a maximum of one group per positive detection using the equation

$$D = \frac{n_s(1-f_s)g}{a_s P_s T_s} \tag{2}$$

(Kyhn *et al.*, 2012), where n_s is the number of detection positive time units, f_s is the false positive rate, a_s is the monitored area size, P_s is the probability of detecting a detection time unit in the monitored area, T_s is the number of detection time units monitored, and g is the mean group size.

The latter formulation is only appropriate for relatively short detection time unit lengths, so that the assumptions of negligible animal movement within the detection time units and single groups per detection hold (Kyhn *et al.*, 2012). Group size estimation is especially challenging for species which, like porpoises, have unstructured, not easily defined groups, and where visual and acoustic group size estimation may differ (Torres Ortiz *et al.*, 2021; Anderson *et al.*, 2023; Rone *et al.*, 2024). To estimate absolute densities from longer detection time units, the mean number of animals detected per detection positive time unit needs to be known, which is usually not available and difficult to estimate. Thus, these are typically used as relative measures to compare diel, seasonal, spatial and pre- vs post-disturbance differences in detections [e.g., Stedt *et al.* (2023) and Voß *et al.* (2023)].

Estimates of P and P_s are sometimes derived experimentally for a subset of sites during limited time periods, by assuming that they are representative and can be extrapolated for different locations and time periods (Kyhn *et al.*, 2012; Nuuttila *et al.*, 2018; Amundin *et al.*, 2022). However, Macaulay *et al.* (2023), highlighted that harbour porpoise behaviour has a substantive impact on P_s, especially foraging, due to variation in turning and click rates. Foraging strategies and rates can vary between individuals and depending on demographic characteristics (Fontaine *et al.*, 2007; Jansen *et al.*, 2012), seasonally, on a diel basis (typically higher foraging rates at night) and between habitats and local sites, even on the order a few hundred meters apart (Schaffeld *et al.*, 2016; Amundin *et al.*, 2022; Macaulay *et al.*, 2023; Stedt *et al.*, 2023; Rojano-Doñate *et al.*, 2024; Hartmann *et al.*, 2025). This would suggest that the assumption that P_s is similar across different spatial and temporal contexts may not hold for porpoises and likely other toothed whale species, as variability in detectability across sites and temporal scales is not necessarily captured.

Macaulay *et al.* (2023) performed simulations assuming the use of a broadband recorder that would detect and classify individual porpoise clicks. A widely used alternative method for PAM is to use automated click detectors [such as the *Cetacean POrpoise Detector* (*C-POD*, Chelonia Ltd.)] instead of a broadband recorder; these devices run an onboard click detector which only detects NBHF-click-like impulses in the 20–160 kHz frequency band and saves meta information on transient detections, vastly decreasing storage requirements (Tregenza, 2013). They come with proprietary software (*CPOD.exe, FPOD.exe*, Chelonia Ltd.) that uses a custom algorithm to group single click detections into click trains (Tregenza, 2013). Click trains are sequences of clicks that are classified into species categories to reduce false positive click detections, since singular false positive signals with porpoise-click-like properties are more likely to occur compared to chance trains that appear to be porpoise trains but originate from a different source (Tregenza, 2013). There is a trade-off between reducing the number of false positives and increasing the number of false negative detections (Sarnocinska *et al.*, 2016; Gauger and Taupp, 2026) and we do not aim to characterise or assess this trade-off. In real-world monitoring data, however, this is a consideration that needs to be made depending on study context (e.g., noise, false positive rates relative to rarity of acoustic detection of the target species) and goals (Caillat *et al.*, 2013; Gauger and Taupp, 2026).

C-PODs and their successor, Full waveform capture PODs (F-PODs), are widely used to estimate porpoise abundance, density, disturbance effects and animal behaviour (Bergès *et al.*, 2020; Amundin *et al.*, 2022; Voß *et al.*, 2023). Detectability estimates vary many-fold, depending on the method of estimating P, click (-train) detection-classification stringency, between sites and depending on click source levels (Sarnocinska *et al.*, 2016; Nuuttila *et al.*, 2018; Amundin *et al.*, 2022; Todd *et al.*, 2025b). It is unclear how the false negative rate, i.e., the proportion of true porpoise click detections rejected by these classifiers, varies by behaviour. This adds an additional layer of behavioural variability in detectability beyond the issue of missed single-click-detections described by Macaulay *et al.* (2023). It is therefore important to understand whether automated click detectors are vulnerable to the same issues around behavioural variability of detectability as broadband recorders (Macaulay *et al.*, 2023).

Our goal is to assess the performance of automated click detectors and click-train classifiers regarding detectability to demonstrate potential unquantified variability in detectability and improve PAM-based inference of absolute abundance. We quantify how behavioural variation affects probability of porpoise click detection, and whether some

11 November 2026 21:14:56

units of detection are less susceptible to this variation than others. We simulate detection data using real-world 3-D harbour porpoise tracks obtained from tagged animals and classified into discrete behavioural states, combined with a propagation model and detection-classification algorithms designed to mimic those used for automated click detectors. We estimate the probability of detecting clicks, click-positive seconds and click-positive minutes, and determine how this varied depending on whether the classification algorithm included click-train classification or not, and how behaviour drove this. Finally, we provide a guideline on how to implement our findings in PAM-studies.

II. MATERIALS AND METHODS

We aimed to quantify how behavioural variation affects the acoustic detection probability of porpoises when using different detection units and classification methods. For this, we developed an analytical framework that integrates tagged-animal dive data, acoustic processing, behavioural classification, and simulated detector performance. The workflow consisted of six main steps: (A) Extracting 3-D dive tracks from tag data, (B) tag acoustic data processing and click-train classification, (C) tag data behavioural state classification, (D) simulating detectors and the detection process, (E) calculating detection probabilities and (F) modelling detection probabilities as a function of behavioural states.

Sections II A and II D were performed in MATLAB (version R2024b, MathWorks, Natick, MA) and Sec. II B in MATLAB (version R2024b, MathWorks, Natick, MA) and *PAMGuard* [Gillespie *et al.* (2026), version 2.02.16]. Section II C was performed in MATLAB (version R2021b, MathWorks, Natick, MA) and Secs. II E and II F in R [version 4.4.3 (R Core Team, 2025)].

A. Extracting 3-D dive tracks from tag data

Ten DTAG-4 tags were deployed on harbour porpoises in the Kattegat and Belt Sea between 2017 and 2023. During commercial fishing operations, porpoises were incidentally and unintentionally entrapped in pound nets by fishermen (Johnson *et al.*, 2009; Macaulay *et al.*, 2023). An expert assessed the porpoises' health, recorded sex, length and age and attached tags via suction cups approximately 5 cm behind the blowhole. Porpoises were released and tags were recovered upon detaching passively after 3–43 h of recording data per tag, encompassing a total of 228 h across all tags. The first hour of each deployment was removed to reduce the behavioural effects of animal handling and entrapment in pound nets according to Rojano-Doñate *et al.* (2018). Tags sampled accelerometer, pressure and tri-axial magnetometer data at 250–625 Hz with 16-bit resolution (Johnson and Tyack, 2003) as well as a GPS position every 1 to 3 min under optimal conditions depending on weather and animal dive cycle. Custom written MATLAB (version R2024b, MathWorks, Natick, MA) scripts and functions from the *AnimalTags (DTAG)* toolbox were used to process DTAG accelerometer, magnetometer and pressure detector data to yield porpoise acceleration ($m\,s^{-1}$), magnetic field (μT) and dive depth (m) at a decimated sampling rate of 25 Hz. The GPS used the snapshot method [similar to FastlocTM (Dujon *et al.*, 2014)] in which a 64 ms acquisition of the demodulated GPS-band radio signal was stored in memory during surfacing. The *ptrack* function from the *AnimalTags* toolbox was used to dead-reckon 3-D paths at an assumed forward speed of $2\,m\,s^{-1}$. The *fit_tracks* function from the same toolbox was used to fit dead-reckoned tracks to the DTAG-4 GPS fixes for more accurate and georeferenced tracks, allowing for estimation of bathymetry and water depth along a track.

B. Tag acoustic data processing and click-train classification

Tags sampled mono audio at 576 kHz. The *PAMGuard* click detector module initially digitally pre-filters and detects signals 10 dB above noise level across a broad frequency range (20–250 kHz high pass Butterworth filter), and then uses a Chebychev bandpass trigger filter that covers the frequency band of porpoise clicks with distortion due to off-axis placement (100–200 kHz). This extracts clicks from the tagged porpoise, other porpoises and false positive clicks not produced by porpoises by comparing signal-to-noise level and extracting those above a detection threshold (10 dB). *PAMGuard's* automated click classifier was used to identify focal clicks produced by the tagged porpoise. These clicks have a peak frequency between 150 and 200 kHz due to frequency distortion when close to the sound source and with the tags being placed far off-axis, making these clicks easy to distinguish from on-axis clicks (Madsen *et al.*, 2010). In the end, three customised classifiers were required to extract clicks produced by the tagged porpoise.

To remove remaining false positive click detections, a clustering algorithm was applied in MATLAB (version R2024b, MathWorks, Natick, MA) (Macaulay *et al.*, 2023). Buzzes defined with inter-click-intervals (ICIs) of <16 ms were removed from the data ahead of the simulation (Wisniewska *et al.*, 2016; Sørensen *et al.*, 2018), as we assume much lower detectability of these clicks due to lower source level and a wider beam pattern (Wisniewska *et al.*, 2015). We estimated precision (98%) and recall (95%) of the final click dataset used in the simulation by manually annotating five randomly selected 2-min segments on each tag and counting false positive, false negative and true positive click detections (see supplementary materials). Apparent output level was calculated as peak-to-peak-amplitude, by subtracting the minimum of a click's waveform from the maximum.

The KERNO-F click-train classifier measures how a set of click characteristics change between clicks, and how inter-click-intervals (ICIs) change (Ivanchikova and Tregenza, 2023). We attempted to imitate the click-train classification process with a simple algorithm to test effects of ICI changes and number of consecutive click detections in different behaviours. A custom click-train classifier was

11 November 2026 21:14:56

https://doi.org/10.1121/10.0047478

written in MATLAB (version R2024b, MathWorks, Natick, MA) and applied to all click detection time series. The recall for different combinations of settings for this classifier was calculated as the proportion of total click detections that were classified into click-trains. Our custom classifier used changes in subsequent ICIs and minimum number of clicks per train to classify trains. This effectively builds click-trains as sequences of detections without gaps, i.e., all clicks produced were detected over the duration of the train (Table 1S). Different combinations of detector settings were tested; minimum number clicks of 3, 4, 5, 6, and 7 clicks per click-train and maximum percentage difference between subsequent ICIs of 2%, 50%, 100%, and 200%. Table 2S shows five studies investigating the recall of automated click detectors, which ranged from 25% to 87%, depending on detection unit, classifier settings, device type used for comparison and experimental setup. We chose 50% maximum percentage increase and decrease between subsequent ICIs and a minimum train length of 5 clicks, which had ~30% recall (Fig. 2S). This choice of settings was based on properties described for KERNO-F and KERNO classifiers (Tregenza, 2013; Ivanchikova and Tregenza, 2023), which has a minimum of 5 clicks per train and a maximum ICI of 250 ms for NBHF clicks (Tregenza, 2013). 50% is a high percentage ICI change when considering regular porpoise trains as they are emitted (75% of percentage ICI changes are below 51.8% in our data, Table 1S), but we saw that click-trains were 'broken up' in our simulated PAM data due to animal movement and beam pattern, leading to only 50% of percentage ICI changes being below 55.5% in PAM detections (Table 1S).

C. Behavioural state classification

Tag data were first segmented into individual dives by identifying the periods between consecutive respirations. To distinguish true dives from short apnoeic submersions, we applied a minimum depth threshold of 3 m and a minimum duration threshold of 20 s. These thresholds were selected based on a preliminary analysis exploring different thresholds of duration and depth on validated data (Rojano-Doñate *et al.*, 2024). Segments that did not meet these criteria were classified as inter-breath-interval apnoeas and excluded from subsequent behavioural classification.

All remaining dives were classified into behavioural states using the Hidden Markov model (HMM) developed by Rojano-Doñate *et al.* (2024). This model assigns dives to one of three states: non-feeding, pelagic feeding, and benthic feeding, based on a suite of kinematic and acoustic variables. The variables used to fit the model were minimum specific acceleration (MSA) as a proxy for activity level, median buzz depth relative to maximum dive depth, pitch and roll circular variance, proportion of time at bottom and surface, circular variance of pitch at the bottom, and presence or absence of buzzes. These behavioural states reflect distinct foraging strategies: Pelagic feeding dives are characterised by buzzes occurring throughout the water column; Benthic feeding dives exhibit buzzes concentrated near maximum dive depth; and non-feeding dives contain few or no buzzes.

D. Simulating detectors and the click detection process

We simulated an array of detectors and the propagation of clicks produced by tagged porpoises to obtain artificial detection time series. This involved the following steps: (1) Simulating detectors around porpoise tracks, (2) estimating porpoise source levels (SL), (3) simulating beam-pattern-dependent sound propagation and energy loss over distance, and (4) estimating uncertainty around SL and the random sampling of tags.

1. Simulated detectors

We simulated an array of detectors surrounding each 3-D track independently. Detectors were placed on a 200 m-spaced horizontal grid extending up to 1200 m from all points along the track, distributed vertically between 5 and 155 m depth (Table I). Because each deployment generated hundreds of thousands to millions of clicks, and the received level of each click had to be calculated for thousands of detector positions, the resulting combinations can result in billions of received-level calculations, which is memory- and time-intensive. To maintain computational efficiency, detector depths were simulated at 10, 30, or 50 m steps between them, depending on the deployment duration and the local maximum bathymetry. Differences in vertical

TABLE I. Summary of tag deployments that were used for the simulation.

Animal ID	Deployment month and year	Tag duration after removal of first hour (h)	N. simulated detectors	Simulated detector depths (m)	Max. bathymetry depth (m)	Max. dive depth (m)	Tag clip level (dB peak-to-peak)
Hp17_135a	May 2017	12.5	6486	5, 15, 25	29	27	173
Hp18_134a	May 2018	42.4	58 470	5, 15, 25, 35, 45, 55, 65, 75, 85	93	80	173
Hp19_262a	September 2019	25.3	6779	5, 15, 25, 35	43	41	173
Hp21_319a	November 2021	3.46	16 307	5, 15, 25, 35, 45, 55, 65, 75, 85	86	85	174
Hp21_330a	November 2021	40.2	5677	5	38	38	174
Hp21_355a	December 2021	16.0	14 268	5, 55, 105, 155	192	98	174
Hp22_296a	October 2022	33.2	13 258	5, 55, 105	153	112	174
Hp23_093a	April 2023	21.9	4384	5, 15	24	23	174
Hp23_104b	April 2023	22.6	17 841	5, 15, 25	31	30	174
Hp23_124a	May 2023	9.76	5792	5, 15	14	14	174

11 November 2026 21:14:56

spacing (10, 30, 50 m) between detector grids were accounted for by including them as a binned covariate in the regression model described in Sec. II F. Bin edges were consistent between animal IDs.

2. Estimating source level

Source level (SL) is defined as the on-axis sound pressure level measured at or back-calculated to 1 m from the animal, in the case of porpoises ahead of the animal (Koblitz *et al.*, 2012). Because the acoustic tags are placed at the back of the head behind the blowhole the output levels recorded on the tag are off-axis SL and the sound energy is lower and distorted (Macaulay *et al.*, 2020). To correct for this, the recorded output level is first converted to an apparent output level (AOL) based on the tag's experimentally derived calibration values (Table I) as

$$AOL = RL + CL, \tag{3}$$

where RL denotes the relative level of each click and, CL the tag clip level. SL was estimated from AOL using an experimentally determined linear relationship (see supplementary materials) between AOL and SL, following Macaulay *et al.* (2023),

$$SL = 0.8866AOL + 41.2728, \tag{4}$$

where AOL is inserted to estimate SL. This equation was applied to each click to derive its estimated SL.

3. Sound energy loss over distance

Once SL was estimated, the click received level (RL) at each simulated detector was estimated by accounting for both transmission loss and directional beam loss

$$RL = SL - BL - TL, \tag{5}$$

where SL denotes estimated source level, the sound pressure level of a click at 1 m distance (dB re 1 μPa pp at 1 m), TL is transmission loss (dB) due to distance-dependent spreading and absorption, and BL denotes beam loss (dB), the loss in sound pressure level due to the vertical and horizontal angles between the porpoise and detector.

Slant distance between the porpoise during click production and each detector were computed from their 3-D positions [latitude (°), longitude (°) and depth (m)]. Transmission loss (TL) was then calculated using a spherical spreading model with a frequency dependent absorption

$$TL = 20\log_{10}(R) + \alpha R, \tag{6}$$

where R denotes distance to the detector (m) and α denotes the absorption coefficient of sound energy (dB/m), estimated at 0.04 dB/m for stereotypical NBHF clicks at 130 kHz peak frequency (Ainslie and McColm, 1998). The rapid attenuation of high frequency soundwaves and lack of data on sediment, salinity and bathymetry made this the most feasible way of estimating TL in the shallow waters of the Kattegat and Belt Sea, and has been shown to be a valid approximation under these conditions (DeRuiter *et al.*, 2010).

Directional attenuation due to the porpoise's beam pattern was simulated based on empirical BL measurements (Macaulay *et al.*, 2020). Using porpoise positions and triaxial orientation, relative angles between the porpoise axis and detectors were calculated and referenced to the BL measurements for respective relative angles to deduct these BL values from received levels.

C-PODs have an estimated detection threshold of approximately 110–117 dB peak-to-peak re 1 μPa at 5 μs resolution (Tregenza, 2013; Dähne *et al.*, 2013; Clausen *et al.*, 2019). We used a constant detection threshold of 116 dB (Clausen *et al.*, 2019). Clicks with predicted RL above that threshold were logged as detections: all other clicks were discarded as non-detections.

4. Estimating uncertainty

To account for error in estimation of SL and simulate random sampling of tags, we propagated two sources of uncertainty via bootstrapping; (1) residual error in the relationship between AOL and SL and (2) random sampling of tag deployments. To simulate the sampling process and estimate uncertainty, we performed a bootstrap of 500 iterations, where on each iteration, ten random tags were sampled with replacement (as our sample size was ten).

Uncertainty in the AOL-SL calibration was estimated using results from pool experiments on two captive porpoises (see supplementary materials). The linear model between AOL and SL included the two porpoises involved in the experiment as random intercept. We extracted the residual errors for each porpoise to the linear equation in the order that click data points were produced by the porpoise. On each bootstrap iteration one of the two porpoises was chosen randomly and the random intercept offset and residual click level errors were added to all SLs according to

$$SL_b = SL + \varepsilon_{click} \pm 0.008, \tag{7}$$

where $\varepsilon_{\text{click}}$ denotes the residual error of a single click data point in the linear model and +/− 0.008 denotes the random intercept of one of the porpoises used to derive the linear equation in a pool experiment (see supplementary materials). Residual errors from the linear model [Eqs. (4), (7)] were added to each click produced by the tagged porpoise in the same sequence in which they were measured in the pool experiment to propagate the temporal autocorrelation in residual error of subsequent clicks. For each tag, the starting point of the first residual click error was chosen randomly.

E. Calculating the detection probabilities

The probability of detection was calculated as the proportion of available detection units that were actually detected within a 0–1200 m radius around every detector. Horizontal distance was calculated as Euclidean distance

11 November 2026 21:14:56

https://doi.org/10.1121/10.0047478

between the 2-D porpoise and detector position [latitude (°) and longitude (°)]. A truncation distance of 1200 m was chosen because tests showed that no clicks were detected beyond this distance. We estimated detection probabilities for three types of detection units: (1) *raw clicks*, each click was treated as a detection unit. Available clicks were all clicks emitted within 1200 m of a detector and detected clicks were those with received level exceeding the detection threshold. (2) *click positive seconds* (CPS), data were binned into 1-s intervals. Available seconds were the number of seconds a porpoise spent within the truncation distance of a detector, and detected units were the number of seconds that contained at least one click detection. (3) *Click positive minutes* (CPM), the same as in seconds but for minutes. For all three detection units, detectability was also estimated for their equivalent click-train-classified version (Table II). For each second and minute, the mean distance from a detector for that second or minute were calculated. Distance to each detector was calculated for each GPS-position sample within a second (25 Hz) and the mean of those distances was taken to get the mean distance for a second. A minute was available if the mean distance of each second within the minute was within the truncation distance.

1. Effective detection area

Detection probabilities become smaller as the truncation distance increases, i.e., the maximum distance for which detectability is estimated. The effective detection area (EDA) is a measure of detectability that standardises monitoring effort. The EDA takes the truncation distance into account to estimate the area within which the same number of detections is missed as were detected outside that area, effectively setting detectability within the EDA to 1 (Kyhn *et al.*, 2012; Marques *et al.*, 2013). In the field, it accounts for environmental variation such as different noise regimes, which was not simulated here (Nuuttila *et al.*, 2018). For all detection units and behavioural state combinations, the effective detection area was calculated following

$$EDA_{ij} = \pi W^2 P_{ij}, \tag{8}$$

where the effective detection area (EDA) and its standard error were calculated using probability of detection P and the maximum truncation distance (W, 1200 m) for every detection unit i and behavioural state j.

2. Behavioural ratios

Ratios of the probability of detection between behavioural states were calculated for all detection units to show differences in detectability as they compare between different detection units. Ratios were calculated using

$$Ratio_{ij} = \frac{P_{ij}}{P_{i1}}, \tag{9}$$

where P_{ij} denotes the probability of detection for detection unit i and behavioural state j, behavioural state 1 being the non-feeding state.

Absolute probabilities are on different scales due to the saturation of positive second and minute detection units, which results in much higher detectability and taking ratios allows for a normalised scale between units.

F. Modelling detection probabilities

We wanted to test the effect of different behavioural, detector and environmental variables on the probability of detection. For modelling purposes and to be able to estimate probabilities of detection by behavioural state, counts of available and detected detection units were obtained for different combinations of binned variables: Horizontal distance from the detector (m), circular variance of heading, roll and pitch, water depth (m), porpoise depth (m), detector depth (m), inter-click-interval duration (s), click source level (dB), day/night (binary), and behavioural state (0–3). For click detections, circular variance parameters were obtained by sampling heading, roll and pitch during the half second in which the click occurred. Bin edges were chosen as evenly spaced for distance, and source level. In cases with more extreme imbalances between evenly spaced levels (circular variance, inter-click-interval and depths), bin edges were chosen by manually examining the distribution of each variable and defining finer bins where the density of data was higher and where we would expect a difference in effect. Bin edges and centres are available in Table 3S.

Two broad types of models were fitted for the three detection units, for click-train-classified and non-train-classified data. The first and full model type (1) that included all behavioural variables was used to examine the effect size of behavioural variables on detectability. The reduced model (2) was used to predict estimates of detectability from the

TABLE II. Terms and abbreviations used to refer to the three different detection units under investigation, and their descriptions, for their respective train-classified and non-train-classified permutations.

	Click	Positive second	Positive minute
Not Train-classified	Click: all detected clicks	Click positive second (CPS): seconds with at least 1 click	Click positive minute (CPM): minutes with at least 1 click
Train-classified	Click within train: all clicks classified as being in click-trains	Train positive second (TPS): seconds with at least 1 click in a click-train	Train positive minute (TPM): minutes with at least 1click in a click-train

11 November 2026 21:14:56

data for each behavioural state. The random effect for animal ID was set to 0 and models were fitted on every bootstrap iteration and mean predicted probabilities and standard errors across bootstrap iterations were then used to obtain estimates of P, EDA, and respective behavioural ratios.

1. Generalised linear models on the probability of detection

We modelled binned behavioural variables for each of the three detection units, for click-train-classified data and non-train-classified data to quantify the effects of different aspects of porpoise behaviour on detectability. A binomial generalised linear mixed effect model (GLMM) with logit link function was fitted using the *bam* function in the *mgcv* package (Wood, 2010) for increased computational efficiency. The response variable was the number of successes (detected units) and failures (available units—detected units). All explanatory variables were binned and included as ordinal categorical variables, except for the binary day/night variable, behavioural states (unordered categorical), and animal ID, which was included as a categorical random effect variable to account for random between-tag variation. Coefficients were extracted for each of the models, variables and bin levels and weighted by prevalence of each bin in the data (proportion of the tag time series spent in each bin relative to total tag duration) to reflect the effective relative effect size of behavioural variables on the probability of detection.

2. Reduced generalised linear models on the probability of detection

We fitted the same models for each detection unit but only including the categorical predictor behavioural state and animal ID random effect. This model was fitted separately for each bootstrap iteration and detection unit and used to predict detection probabilities for each bootstrap iteration, behavioural state and detection unit type. This method preserved variation in detectability due to behavioural variables, since we removed those explanatory variables from this model (apart from the behavioural state categorical predictor itself to allow for making predictions). Using this method, we also propagate the residual error in the relationship between AOL and SL and account for the individual differences between animals and tag deployments, due to tag placement, individual variation and deployment duration.

III. RESULTS

A. Behavioural states

228 h of tag data were classified into 57.8 h (25% of time) of non-feeding, 71.4 h (31% of time) of inter-breath-intervals (i.e., near the surface), 66.6 h (29% of time) of pelagic feeding and 33.1 h (15% of time) of benthic feeding (Fig. 1). Inter-breath-interval is not a behavioural state in the HMM analysis but is retained as a separate behavioural state to investigate in our detectability analysis. A mean of 25 123 clicks h^{-1} [95%-confidence interval (C.I.) 21 284–28 962 clicks h^{-1}] was produced, with 13 336 clicks h^{-1} [C.I. 11 057–15 615 clicks h^{-1}] during non-feeding behaviour, 18 097 clicks h^{-1} [C.I. 14 033–22 160 clicks h^{-1}] during inter-breath-intervals, 34 126 clicks h^{-1} [C.I. 25 409–42 842 clicks h^{-1}] during benthic feeding and 34 943 clicks h^{-1} [C.I. 29 200–40 668 clicks h^{-1}] during pelagic feeding. Buzz presence (<16 ms ICI) was highest during benthic feeding (mean 22% of seconds) and lowest during non-feeding behaviours (mean 4% of seconds) with a mean of 14% buzz positive seconds, which is expected as buzzes were used during the HMM classification. Mean dive depth and bathymetry varied considerably between individuals, with some porpoises diving to a maximum depth of 112 m at 153 m maximum water depth and others remaining in water depths of a maximum of 14 m and diving to a maximum of 14 m. Porpoises did not dive beyond 112 m, meaning they did not dive to the bottom where the bathymetry exceeded this (two tags, Table I). Mean dive depth was 8 m [C.I. 6–10 m], with 6 m [C.I. 4–7 m] during non-feeding behaviour, 1 m [C.I. 0–1 m] during inter-breath-intervals, 14 m [C.I. 9–19 m] during pelagic feeding and 18 m [C.I. 10–26 m] during benthic feeding. Click source level varied considerably between tags, with a mean of 170 dB [C.I. 163–179 dB] across tags and a range of 153–187 dB mean source level between tags, and between behaviours (mean of 171 dB [C.I. 166–178 dB] during pelagic feeding, 170 dB [C.I. 161–177 dB] during non-feeding and 162 dB [C.I. 150–169 dB] during benthic feeding, i.e., 9 dB lower during benthic feeding). Mean circular variance of heading, roll and pitch was highest during pelagic feeding (0.13 [C.I. 0.11–0.15], 0.06 [C.I. 0.04–0.07], 0.02 [C.I. 0.0–0.02]). During benthic feeding, mean circular variance of heading, roll and pitch was second highest (0.12 [C.I. 0.09–0.17], 0.04 [C.I. 0.03–0.06], 0.01 [C.I. 0.01–0.02]). Pitch and roll were used in the HMM, explaining this, but heading was not.

B. Click-train classification

Recall of the click-train classifier ranges from 1% for at least 7 clicks per train and 2% maximum percentage subsequent ICI change to 75% for at least 3 clicks per train and 200% maximum percentage subsequent ICI change (Fig. 2S). For the results presented here, we chose 50% maximum percentage increase and decrease between subsequent ICIs and a click-train length of at least five clicks, according to the properties described for KERNO and KERNO-F (Tregenza, 2013; Ivanchikova and Tregenza, 2023). Recall is lowest for benthic feeding clicks (21%) and highest for non-feeding clicks (28%) (Fig. 3S). Pelagic feeding has a recall of 25% (Fig. 2S).

Relative detectability of different behaviours shows small differences between most of the different click-train settings tested (Fig. 2S). However, especially for very low recall, the probability of detecting different behaviours converges, especially in train positive seconds and minutes.

11 November 2026 21:14:56

https://doi.org/10.1121/10.0047478

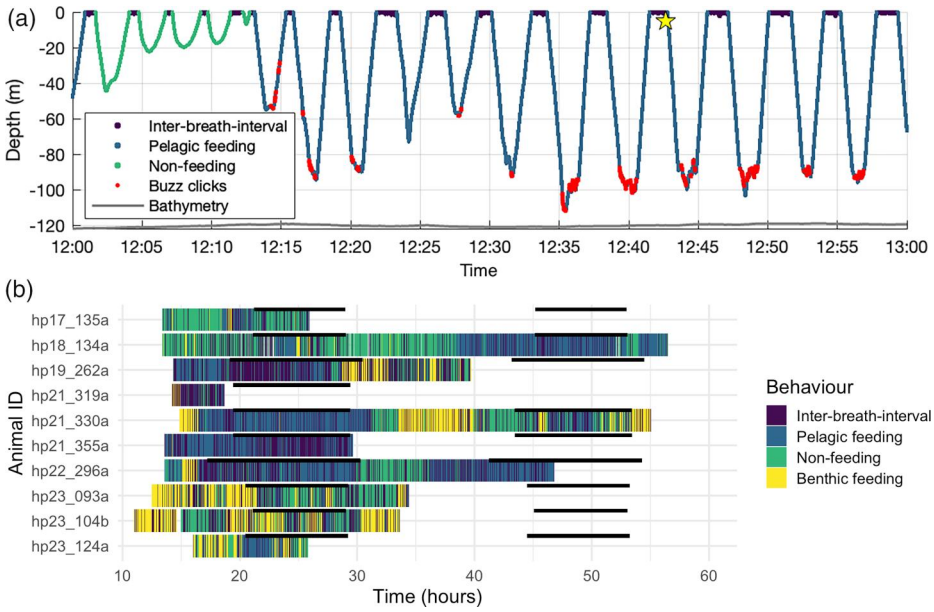


FIG. 1. (a) Example of an hour of dive track with time on the *x* axis and dive depth on the *y* axis, coloured by behavioural states included in the Hidden Markov model (except for inter-breath-intervals, which are short periods spent at the water surface that are not classified in the HMM as a separate behavioural state). Red markers indicate buzz clicks, where inter-click-interval was below 16 ms. The star marks the position of an example detector the porpoise passed at 5 m depth. (b) Tag deployment periods as hour of the day (*x* axis), coloured by behavioural state classification, illustrating differences in deployment duration and time spent in behavioural states between tag deployments. Black bars indicate nighttime.

C. Probability of detection

To standardise effort, we express detectability as EDA (see supplementary materials Table 3S and Table 4S for estimates of P and EDR, EDA). The mean detectability, and thus EDA, of CPM is the highest, followed by TPM, CPS and TPS (Fig. 2). Click-train-classified data shows lower detectability than non-click-train-classified counterparts as the click-train classifier discards true clicks that were not assigned to a click-train (Fig. 2).

1. EDA

Figure 2 shows the EDA and EDR by detection unit and behavioural state, with the size of slices being weighted by the proportion of hours spent in each state across tags. The EDA is 28 times larger for CPM compared to click data and 4 times larger for CPS. In click and train data, pelagic and benthic feeding is smaller in EDA compared to non-feeding behaviours (0.75 and 0.79 the size of non-feeding states in pelagic feeding). The EDA of pelagic feeding is 1.69 and 1.61 that of non-feeding behavioural states for TPS and CPS and 1.78 and 1.61 in TPM and CPM, meaning detectability of foraging increases both in absolute terms and relative to non-feeding in these detection units.

2. Behavioural ratios

Pelagic feeding is the most different from the non-feeding state in TPM, where it is twice as detectable as non-feeding (Fig. 3). In click and train data, non-feeding is twice as detectable as benthic feeding. Feeding states become more detectable relative to non-feeding in CPS, TPS, CPM, and TPM (Fig. 3). Uncertainty is higher in train-classified data compared to non-train-classified across all detection units (Fig. 3).

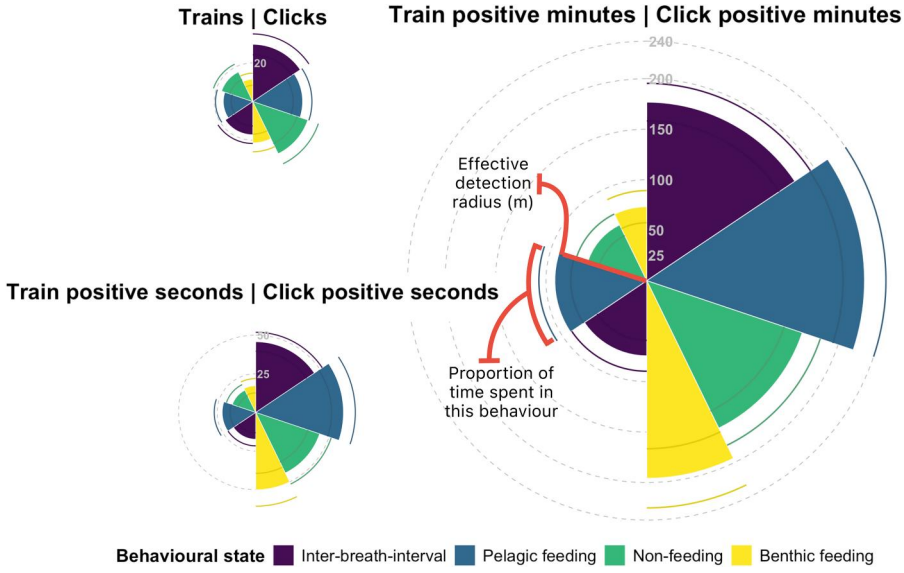


FIG. 2. Effective detection area (EDA) and radius in meters of each detection unit (3 circles) where the left half is click-train classified (trains, TPS and TPM) data and the right half non-click-train-classified data (clicks, CPS and CPM) with standard errors. Each coloured slice indicates the EDA of a different behavioural state, and their relative central angle (i.e., the angular size of each slice) reflects the proportion of hours spent in each behavioural state across all the tag data. The grid ticks indicate the radius in meters. Note that click-train-classified EDAs on the left half are smaller due to low classifier recall reducing the overall number of detections, and therefore detectability.

11 November 2026 21:14:56

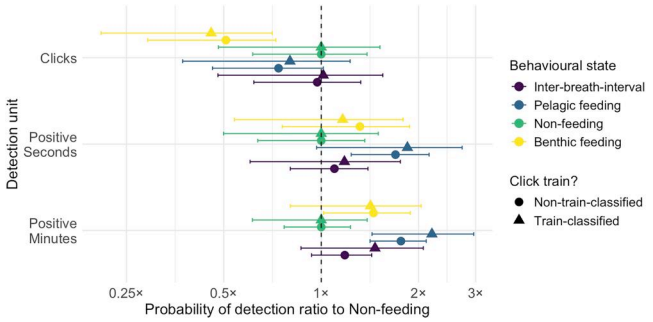


FIG. 3. Predicted mean probabilities of detection across bootstrapped model predictions as a ratio of that of the non-feeding state. Uncertainty represents the standard error in the probability of detection across bootstrap iterations of model predictions for each behaviour, as a ratio of the mean non-feeding detectability estimate. Data points are coloured by behavioural state, and triangles indicate click-train-classified data. The line at 1 indicates the point where a probability would be the same as the mean of the non-feeding state. Uncertainty on non-feeding is the ratio relative to its own mean.

Table III shows "worst case scenarios," given detectability is measured for one behaviour but porpoises exhibit exclusively another behaviour which is the most different from the first behaviour at a site. Our model predicts similar ratios for this across detection units in non-train-classified data: between a 1.42-fold difference to a 3.51-fold difference in detectability between the most and least detectable behaviour. In train-classified data, this range increases to 1.23- to a 3.87-fold difference, with an increase in this ratio from trains to TPS to TPM, so longer detection time units appear to exacerbate differences between behaviours in detectability and uncertainty in this ratio.

D. Drivers of the probability of detection

Figure 4 shows the coefficient size of the predictors with the biggest relative effects on the probability of detection, weighted by the prevalence of each bin in the data, including time of day, distance, ICI as a proxy for click rate, detector depth, behavioural state and circular variance of heading. Click rate, click source level, distance and dive behaviour (circular variance of heading, depth) appear to be the predictors with the biggest effect sizes, especially for CPS, TPS, CPM, and TPM. The effect of click rate and circular variance diminishes in click data. For every unit, its train-classified equivalent shows a more negative effect of longer ICI. Similarly, increasing heading variance has a positive effect in non-train data and TPM and a negative effect in click-trains and TPS. Daylight has a negative effect and more shallow detectors a positive effect on the probability of detection. These behavioural and detector variables explain 80%–92% of deviance in these models.

TABLE III. The mean, 2.5th and 97.5th quantile of the ratios between the highest detectability behaviour to the lowest detectability behaviour across bootstrap iterations for each detection unit, illustrating the worst case of P being estimated during one behaviour and applied at a site with another behaviour.

Detection unit		Mean	Lower bound	Upper bound
Non-Train	Clicks	2.13	1.42	3.11
	CPS	2.09	1.53	3.51
	CPM	1.9	1.49	2.93
Train	Trains	2.46	1.23	3.71
	TPS	2.25	1.66	3.82
	TPM	2.36	1.81	3.87

IV. DISCUSSION

Using a dataset of 228 h of harbour porpoise tag data, we estimated the probability of detecting clicks, CPS, TPS, CPM, and TPM, and tested the effect of click-train classification and porpoise behaviour on that probability, by simulating automated click detectors as model-devices. We find similar issues with variability in detectability between behaviours in click-train-classified data and when using detector settings to mimic automated click detectors, such as C-PODs. Depending on detector-type-specific characteristics, the problems we see likely apply to a range of detectors, although perhaps in different ways, e.g., detection thresholds and classifier performance may affect behavioural variation in detectability (Macaulay *et al.*, 2023).

A. Variation in detectability

Across detection time units (CPS, TPS, CPM, and TPM), pelagic feeding is 1.5–2 times as detectable as non-feeding states. Higher click rates and turning rates during feeding increase the effective volume ensonified across a time unit, with more detectors being scanned across in a second or minute and thus the probability of detection is greater, an effect previously shown in broadband detectors (Macaulay *et al.*, 2023). Our model results corroborate this effect, with heading variance having a positive effect on detectability in CPS, CPM, and TPM. Detectability of single clicks depends on source level, distance from the detector and depth, but not turning or click rates (Fig. 4) (because the instantaneous beam pattern during a click is constant and thus the ensonified volume is the same regardless of turning rate). Given the click rate is known, detectability is not affected by it, as it is accounted for when deriving monitoring effort, expressed as a number of clicks available for detection, which scales with click rate. In detection positive time units, the availability denotes the time porpoises spend within the truncation radius, so higher click rates (as seen during feeding) increase detectability, which is an effect we demonstrate (Fig. 3). Improved detectability at shallower depths has been demonstrated in C-POD field studies and we also see a strong positive effect of shallower depths across detection units (Alonso and Nuuttila, 2015). This might be caused by porpoises spending more time close to the surface, e.g., to take breaths, and by more time spent in horizontal orientation resulting in less beam loss.

Click-train classification increases detectability of pelagic feeding relative to other states but decreases it for benthic feeding. We hypothesise that a click-train classifier misses detections where the porpoise turns at a high rate, and thus only ensonifies a detector for long enough for a few

11 November 2026 21:14:56

https://doi.org/10.1121/10.0047478

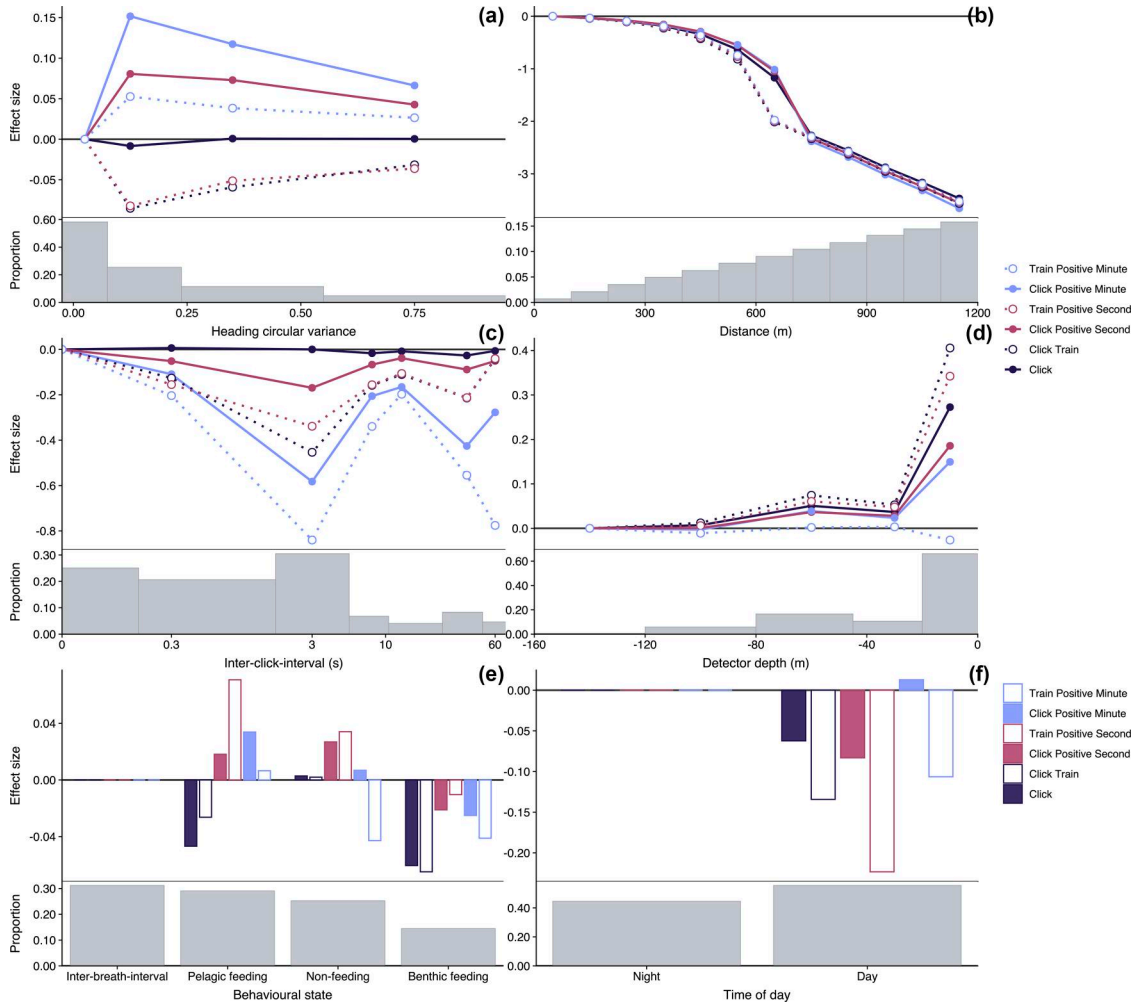


FIG. 4. Effect size of predictors (y axis) in the GLMM modelling probability of detection plotted against predictor size or category in 6 models: train positive minutes, click positive minutes, train positive seconds, click positive seconds, train detections, click detections. Predictors (a) heading circular variance, (b) distance (m), (c) inter-click-interval (s), and (d) detector depth (m) are continuous but binned variables and (e) behavioural state and (f) time of day are categorical, where Night falls between sunset and sunrise and Day falls between sunrise and sunset. Proportion is the effect weight (i.e., the time porpoises spent in each of these bins proportional to the total deployment period) and is indicated by the relative size of the gray bars for each bin. Where distances between bin centres are not spaced evenly, the width of the bars indicates this. Dotted lines and unfilled points/bars indicate train-classified data, whereas solid symbols are unclassified data. Each predictor has a reference level at $y = 0$, which can be seen by all continuous plots beginning at $y = 0$ and the categorical panels (e) and (f) have e blank reference category. Points indicate the value of bin centres on the x axis and the coefficient on the y axis, i.e., a negative coefficient represents a negative effect of that predictor value on the probability of detection relative to the reference level, and a positive coefficient represents a positive effect on the probability of detection. For example, increasing the distance to a detector relative to distance = 0 has a negative effect on the probability of detection.

clicks to be detected consecutively. Effectively, click-trains are "broken up" between detectors due to rapid turning. We see evidence of this in our model results, where train and TPS detectability is in a negative relationship with heading variance. In pelagic feeding, the rapid turning may be compensated by rapid click sequences, so that a short time of continuous ensonification is enough for click-trains to be captured. This suggests that the increase in click rate is roughly proportional to the increase in turning rate, but with slight difference in this proportionality between pelagic and benthic feeding. Click-train classification exacerbates the increase in detectability of pelagic feeding in TPM as high click rates increase the chances of click-trains being captured, and rapid turning means that across a minute, multiple sensors capture click-trains. Our model results provide support for this, with ICI (a proxy for click rate) having a negative effect across detection units, but each click-train-classified unit equivalent showing a more negative effect size than the non-train-classified counterpart. Click rate thereby becomes a bigger driver of detectability in train-classified contexts. Benthic feeding is more variable in terms of click rate, turning rate, dive depths and source levels, perhaps due to adjustments to avoid echoes, clutter and reflections from the sea floor (Wisniewska *et al.*, 2015). This higher behavioural variability, deeper dive depths and pitching the beam pattern towards the floor might negatively affect the probability of capturing enough consecutive clicks to capture click-trains and might also explain non-feeding being twice as detectable as benthic feeding in click and train data (Fig. 3). Click-train classification may also favour

11 November 2026 21:14:56

detection sequences at closer distances (Figs. 13S, 16S), as the probability of a detector being ensonified continuously to capture multiple clicks consecutively is increased when the porpoise is closer. Differences in click-train classifier recall support this hypothesis, as benthic feeding had the lowest recall after click-train classification.

B. Implications for passive acoustic monitoring

Relative differences in detections are often used to infer animal abundance, behaviour or activity, e.g., to measure the behavioural response to noise from pile-driving or investigate spatiotemporal feeding behaviour (Brandt *et al.*, 2011; Benjamins *et al.*, 2017; Stedt *et al.*, 2023; Stavenow Jerremalm *et al.*, 2025). It is often unclear if the spatiotemporal differences in the number of detections are a product of changing animal abundance or behaviour, e.g., switching between feeding and non-feeding states. Experimentally estimated probabilities of detection [e.g., via tracking animals using hydrophone arrays and playback experiments (Amundin *et al.*, 2022)] and click rate on a small subset of sites, detectors and contexts may not always be sufficient to capture and average out detectability and click rates for an entire study. Estimates of detection probability may not necessarily be representative of a study area and period and transferring and extrapolating estimates of detectability may lead to under- or over-estimates of animal abundance and activity.

Porpoises produce more clicks per unit time when feeding. Therefore, the absolute number of click detections may still be higher when a porpoise is feeding around a detector, given similar click detectability. Click rates, source level and dive behaviour have been shown to vary considerably depending on, e.g., foraging strategies, with foraging buzz detectability being lower due to lower source level (Linnenschmidt *et al.*, 2013; Macaulay *et al.*, 2023; Rojano-Doñate *et al.*, 2024; Todd *et al.*, 2025b). Therefore, for density estimation, accurate estimates of click rates and how they vary between sites and over time due to animal behaviour are required when using clicks as a detection unit, unless the resolution of the analysis is coarse enough for behavioural effects to average out across time and space.

More click detections are commonly reported at night (Carlström, 2006; Todd *et al.*, 2009; Osiecka *et al.*, 2020), likely reflecting high nocturnal feeding rates, particularly pelagic feeding in open waters (Schaffeld *et al.*, 2016; Wisniewska *et al.*, 2016; Macaulay *et al.*, 2023; Stedt *et al.*, 2023; Rojano-Doñate *et al.*, 2024). In keeping with that, we find a negative effect of day on detectability across detection units. Temporally varying proportions of time spent feeding on a diel, monthly and seasonal scale may cause unaccounted for variability in detectability within a site over time (Todd *et al.*, 2009; Schaffeld *et al.*, 2016; Kastelein *et al.*, 2018; Stedt *et al.*, 2023). Interpreting this variability as a change in density or distribution over time may be inaccurate.

Beyond ecological drivers of behavioural state, porpoises have been shown to increase mean source levels by 7–17 dB in response to high frequency vessel noise (Hermannsen *et al.*, 2025) and reverberation (Ladegaard and Madsen, 2019). On the other hand, if noise causes porpoises to cease foraging (Wisniewska *et al.*, 2018), and decrease click output (Elmegaard *et al.*, 2023), detectability may decrease. If porpoises change their behaviour throughout the study period or between sites, either due to ecological factors or induced by the noise (Wisniewska *et al.*, 2018; Hermannsen *et al.*, 2025), discriminating behavioural changes from distribution changes is difficult [e.g., to study the effects of noise (Brandt *et al.*, 2011; Tollit *et al.*, 2019)].

Higher detection thresholds, as seen in automated click detectors (Clausen *et al.*, 2019) decrease detectability of low-source-level states such as benthic feeding more than for other states. This may explain our finding of substantially lower benthic feeding detectability compared to Macaulay *et al.* (2023), who used a detection threshold 16 dB lower to reflect broadband detectors. Therefore, placing automated click detectors in areas where porpoises tend to feed benthically will likely have a different effect on detectability compared to a broadband detector. Porpoises exhibit benthic feeding in shallow coastal waters, especially during the day and then move into deeper open waters opportunistically for pelagic feeding at night (Fontaine *et al.*, 2007; Schaffeld *et al.*, 2016; van Beest *et al.*, 2018; Stedt *et al.*, 2024; Rojano-Doñate *et al.*, 2024). This implies that differences in detectability between behavioural states may show a systematic pattern, where detectors in deeper open waters have a higher probability of detection than those in shallower waters due to the prevalence of different foraging strategies and depending on prey availability. While we see similar trends in detectability to Macaulay *et al.* (2023), these differences show that detector-specific characteristics such as detection thresholds and click (-train) detection-classification likely interacts with detectability differences between behaviours. C-POD recall estimates are highly variable, between 21 – 94% with poor correlation with broadband detections, which may hint at underlying variability in the detection-classification process between detector types, but also varying noise regimes that produce different false positive rates (Sarnocinska *et al.*, 2016). We saw increased uncertainty in detectability in click-train-classified detections across detection units, which further hints at this unaccounted for variability in detector performance and variability in false negative rates within and between behaviours. The effects of click-train classification are highly dependent on behavioural state, with benthic and pelagic feeding becoming more different in detectability compared to non-train data due to the interaction of click rates and turning rates. We see that our click-train classifier had the lowest recall during benthic feeding, which supports the idea that benthic feeding is affected differently from pelagic and non-feeding due to behavioural variability. It may be the case that real-world click-train classifiers have different recall rates depending on behaviour, which can exacerbate detectability differences.

11 November 2026 21:14:56

The overall detectability of time-based units is substantially higher (4 × for positive seconds, 28 × for positive minutes), which needs to be accounted for when estimating abundance with this type of data. Detection time units saturate, especially as they increase in length. This increases the probability of detection, as it only requires one click detection for, e.g., an entire minute to count as detection positive. Animal movement across a detection time unit also means that multiple detectors may be ensonified during one time unit, further increasing detectability. These effects mean that the linear relationship between the number of detections and density does not hold.

C. Advice for PAM users and future work

While we find that click-train classification produces varying false negative rates between behaviours and increased variability in detection probability, this should not be to discourage the collection and use of these data. Where false positive detection rates are high and/or difficult to estimate, it may not always be possible to effectively identify false positive clicks based on acoustic properties of single impulses alone (e.g., in the case of some sediment noise and click-echoes), and in such cases a click-train classifier can be effective at identifying true porpoise clicks in various noise regimes (Sarnocinska *et al.*, 2016). Our findings should serve to demonstrate potential issues with false negative rates of different behaviours and how they can be addressed to improve inference from automated click detector data.

Spatial and seasonal trends in foraging strategies due to, e.g., prey distribution may not always average out, and many studies seek to investigate small-scale, or pre- and post-disturbance density on a short temporal scale (Sveegaard *et al.*, 2012; Schaffeld *et al.*, 2016; van Beest *et al.*, 2018; Stedt *et al.*, 2023; Todd *et al.*, 2025a). In such cases, relative and absolute density estimates need to account for uncertainty due to behaviour. For detectability, we provide worst-case detectability differences for the detection units we investigated. Click-train classification appears to introduce additional uncertainty around detectability, with worst-case detectability differences ranging from 1.23 × to 3.87 × differences between behaviours (Table III). Train positive minutes have a 1.81 × to 3.87 × worst-case difference in detectability between the two most different behaviours, which is the highest difference. This uncertainty should be propagated when estimating density based on train-classified summary detection units. With our simulation approach, it may be possible to do further work on how click and click-train characteristics such as ICI, click rates and received level translates from the porpoise tag data (our assumed ground-truth) to PAM detector data and if it is possible to estimate, e.g., accurate click rates from monitoring data. Click rates can be used to account for some of the variability we see and could be estimated directly using our tag data.

Experimentally estimated detectability could use visual data during experiments to determine behavioural-state-specific detectability, which can be used to propagate uncertainty into density estimates [e.g., drone behavioural classification (Stedt *et al.*, 2024)]. When using click-train-classified data, it may not be sufficient to identify foraging and non-foraging behaviours to account for detectability differences, as depending on click rate and turning rate, the effect of benthic and pelagic feeding on detectability may be very different. As porpoises tend to forage benthically in shallower areas where the seabed is accessible (Fontaine *et al.*, 2007; van Beest *et al.*, 2018; Rojano-Doñate *et al.*, 2024), the bathymetry of a study area and differences between sites may affect the variability in detectability due to behaviour and click-train classification.

Future work may involve detection of behavioural states in PAM data directly by developing a behavioural classifier to adjust for state-specific detectability. Developing a behavioural classifier would be valuable, but validation would require sufficiently large datasets containing concurrent fine-scale behavioural observations (e.g., tag data) and PAM detections. To our knowledge, such datasets are not currently available, although they could be collected experimentally (Tyack *et al.*, 2011). Using such a classifier, even in cases where an absolute probability of detection or click rate cannot be calculated, knowing the ratios in detectability between different behavioural states in PAM data can reduce variability of relative abundance estimates, allowing for more accurate comparison between sites and over time. However, a behavioural classifier likely cannot be generalised across or within populations (e.g., in the North Sea) and would need to be tested across spatial and temporal settings to be applicable across studies.

D. Limitations

Our simulation assumes no variation in the effect of noise, which is affected by currents, tides, anthropogenic activity and other site features, and noise has been shown to mask echolocation clicks and affect detectability and behaviour (Clausen *et al.*, 2019; Hermannsen *et al.*, 2025; Veneruso *et al.*, 2025). Transmission loss for highly directional on-axis NBHF echolocation clicks can be approximated according to spherical spreading due to high sound attenuation rates, and propagation and transmission loss. However, transmission loss and thus detectability has been shown to vary (±10 dB, DeRuiter *et al.*, 2010) between sites because of long- and short-term environmental fluctuations and seafloor features (Nuuttila *et al.*, 2018; Amundin *et al.*, 2022). Estimating this effect requires location-specific environmental data we did not have access to or simulate. Beyond these external factors, detection thresholds of individual detectors reportedly vary by ±3 dB in C-PODs (Dähne *et al.*, 2013) and range between 123–132 dB in T-PODs (Kyhn *et al.*, 2008), which we did not account for in our uncertainty and error estimation, where we assumed no differences between detectors or within detectors over time (Clausen *et al.*, 2019).

11 November 2026 21:14:56

Beam pattern variability (Koblitz *et al.*, 2012; Wisniewska *et al.*, 2015; Macaulay *et al.*, 2020) was not accounted for, and only a constant empirical average beam pattern was used in the simulation. Buzzes were therefore removed as they are characterised by a widened beam pattern and much lower detectability. Given that the high directionality of the porpoise beam pattern drives variability in detectability of behavioural states, variability of the beam pattern may affect detectability estimates which we did not account for. Side lobes on the porpoise beam pattern and off-axis energy can result in frequency-distorted click detections, making up 5–50% of click detections in PAM data depending on detection thresholds, which was not incorporated into the simulation (Macaulay *et al.*, 2020).

It is unlikely that 10 tag deployments reflect the full behavioural variability of porpoises, especially considering context-dependent behaviours under different environmental and demographic conditions (Fontaine *et al.*, 2007; Jansen *et al.*, 2012; Schaffeld *et al.*, 2016). We were only able to simulate single porpoises, as tag data of multiple individuals in a group is not available, so the effects of multiple porpoises clicking near a detector were not investigated.

As we do not have access to the C-POD KERNO click classifier algorithm, we did not compare the performance of our click-train classifier to KERNO and it is possible that KERNO affects detectability differently from our classifier. The simulation does not contain false positive click detections, i.e., every click detection is a true porpoise click. Therefore, all click detections filtered out by our custom click-train classifier are false negative detections. This likely does not hold true for the KERNO classifier, as false positives are guaranteed to occur in real monitoring data, and we do not assess the ability of a classifier to identify false positives and the trade-off with false negative rates (Roberts and Read, 2015). Our work must therefore be viewed as a study of false negative rates of a click-train classifier during different behavioural states.

V. CONCLUSION

Variability of detection probability poses a substantive problem that may result in inaccurate and/or imprecise inference on harbour porpoise abundance, density and disturbance effects. Click-train classification may exacerbate detectability differences, both between different feeding states and between pelagic feeding and non-feeding. The limitations of PAM methodology and inference need to be communicated and accounted for by (a) collecting information on study area- and/or site-specific porpoise behaviour, ecology and habitat, especially during experiments to estimate detectability and/or (b) propagating uncertainty due to behaviour and detectability into density estimates. We present, as a reference point for current PAM studies, the potential uncertainty that density estimates are subject to due to behavioural variation in different detection-classification methods. In future work, we hope to classify behaviour in PAM data, thereby allowing for adjustment of detectability in density estimation.

SUPPLEMENTARY MATERIAL

See the supplementary material for details on the PAMGuard click detection and classification process, validation of the simulation, tag calibration, source level estimation and additional exploratory results.

ACKNOWLEDGMENTS

Thank you to Emily T. Griffiths, Charlotte Rose Findlay, Elisabeth DeBusschere, Stijn Vermaere, Jakob Tougaard, Mia Lybkaer Kronborg Nielsen and Ellen Rose Jacobs, for support as part of the DTO-Bioflow project, under which this work has been funded and for helping with logistics around sending data. We thank Freja Jakobsen, Kirstin Anderson, the team at Fjord and Baelt and Chloe Malinka (University of St Andrews) for collecting auxiliary data to calibrate DTAGs. Thank you to Nick Tregenza for providing comments on the manuscript and insightful discussions around automated click detectors. This study was funded by the European Union under the Horizon Europe Programme with Grant Agreement No. 101112823 (DTO-BioFlow). Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union or the European Climate, Infrastructure and Environment Executive Agency (CINEA). Neither the European Union nor the granting authority can be held responsible for them. Part of the equipment was funded by the German Federal Agency of Nature Conservation.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

Data collection was carried out under permission issued to Jonas Teilmann as follows: for the period 2012–2023 by the Danish Forest and Nature Agency (SNS-342-00042); for 2012–2014 by the Animal Welfare Division (Danish Ministry of Justice, 2010/561–1801); for the period 2015–2019 by the Environmental Protection Agency (Ministry of Environment and Food of Denmark, NST-3446-0016) and the Animal Experiments Inspectorate (Ministry of Environment and Food of Denmark, 2015-15-0201–00549). Ethics approval was obtained from the ethics committee of the School of Biology, University of St Andrews (Approval code BL18570).

DATA AVAILABILITY

A 12-h subset of harbour porpoise tag data and associated processing code to simulate sensors and obtain detection probabilities for different acoustic detection units is available. The data that support the findings of this study are

11 November 2026 21:14:56

openly available in hp-detectability at https://github.com/dinaurh/hp-detectability, reference number [doi 10.5281/zenodo.23267153].

Ainslie, M. A., and McColm, J. G. (**1998**). "A simplified formula for viscous and chemical absorption in sea water," J. Acoust. Soc. Am. **103**, 1671–1672.

Alonso, M. S., and Nuuttila, H. K. (**2015**). "Detection rates of wild harbour porpoises and bottlenose dolphins using static acoustic click loggers vary with depth," Bioacoustics **24**, 101–110.

Amundin, M., Carlström, J., Thomas, L., Carlén, I., Koblitz, J., Teilmann, J., Tougaard, J., Tregenza, N., Wennerberg, D., Loisa, O., Brundiers, K., Kosecka, M., Kyhn, L., Ljungqvist, C. T., Sveegard, S., Burt, L. M., Pawliczka, I., Jussi, I., Koza, R., Arciszewski, B., Galatius, A., Jabbusch, M., Laaksonlaita, J., Lyytinen, S., Niemi, J., Šaškov, A., Macaulay, J., Wright, A. J., Gallus, A., Blankett, P., Dähne, M., Acevedo-Gutiérrez, A., and Benke, H. (**2022**). "Estimating the abundance of the critically endangered Baltic Proper harbour porpoise (*Phocoena phocoena*) population using passive acoustic monitoring," Ecol. Evol. **12**, e8554.

Anderson, D., Shuster, L., Elliser, C. R., MacIver, K., Gless, E. J., Krieger, J., and Hall, A. (**2023**). "Harbor porpoise aggregations in the Salish Sea," Oceans **4**, 269–285.

Benjamins, S., van Geel, N., Hastie, G., Elliott, J., and Wilson, B. (**2017**). "Harbour porpoise distribution can vary at small spatiotemporal scales in energetic habitats," Deep Sea Res. Part II: Topical Stud. Oceanogr.**141**, 191–202.

Bergès, B. J. P., Geelhoed, S., Scheidat, M., and Tougaard, J. (**2020**). "Quantifying harbour porpoise foraging behaviour in CPOD data," Proc. Mtgs. Acoust. **37**, 070008.

Brandt, M., Diederichs, A., Betke, K., and Nehls, G. (**2011**). "Responses of harbour porpoises to pile driving at the Horns Rev II offshore wind farm in the Danish North Sea," Mar. Ecol. Prog. Ser. **421**, 205–216.

Caillat, M., Thomas, L., and Gillespie, D. (**2013**). "The effects of acoustic misclassification on cetacean species abundance estimation," J. Acoust. Soc. Am. **134**, 2469–2476.

Calonge, A., Navarro González, P., Mortelmans, J., Deneudt, K., Hernandez, F., and Debusschere, E. (**2024**). "Cetacean passive acoustic network in the Belgian part of the North sea," Sci. Data **11**, 971.

Carlström, J. (**2006**). "Diel variation in echolocation behavior of wild harbor porpoises," Mar. Mammal Sci. **21**, 1–12.

Clausen, K. T., Tougaard, J., Carstensen, J., Delefosse, M., and Teilmann, J. (**2019**). "Noise affects porpoise click detections – the magnitude of the effect depends on logger type and detection filter settings," Bioacoustics **28**, 443–458.

Dähne, M., Katharina Verfuß, U., Brandecker, A., Siebert, U., and Benke, H. (**2013**). "Methodology and results of calibration of tonal click detectors for small odontocetes (C-PODs)a)," J. Acoust. Soc. Am. **134**, 2514–2522.

DeRuiter, S. L., Bahr, A., Blanchet, M.-A., Hansen, S. F., Kristensen, J. H., Madsen, P. T., Tyack, P. L., and Wahlberg, M. (**2009**). "Acoustic behaviour of echolocating porpoises during prey capture," J. Exp. Biol. **212**, 3100–3107.

DeRuiter, S. L., Hansen, M., Koopman, H. N., Westgate, A. J., Tyack, P. L., and Madsen, P. T. (**2010**). "Propagation of narrow-band-high-frequency clicks: Measured and modeled transmission loss of porpoise-like clicks in porpoise habitats," J. Acoust. Soc. Am. **127**, 560–567.

Dujon, A., Lindstrom, T., and Hays, G. (**2014**). "The accuracy of Fastloc-GPS locations and implications for animal tracking," Methods Ecol. Evol. **5**, 1162–1169.

Elmegaard, S. L., Teilmann, J., Rojano-Doñate, L., Brennecke, D., Mikkelsen, L., Balle, J. D., Gosewinkel, U., Kyhn, L., Tønnesen, P., Wahlberg, M., Ruser, A., Siebert, U., and Madsen, P. (**2023**). "Wild harbour porpoises startle and flee at low received levels from acoustic harassment device," Sci. Rep. **13**, 16691.

Fontaine, M. C., Tolley, K. A., Siebert, U., Gobert, S., Lepoint, G., Bouquegneau, J.-M., and Das, K. (**2007**). "Long-term feeding ecology and habitat use in harbour porpoises *Phocoena phocoena* from Scandinavian waters inferred from trace elements and stable isotopes," BMC Ecol. **7**, 1.

Gauger, M. F. W., and Taupp, T. (**2026**). "Harbor porpoises and the machine: Assisting manual validation of click trains recorded by Cetacean Porpoise Detector," J. Acoust. Soc. Am. **159**, 632–646.

Gillespie, D., Macaulay, J., Oswald, M., and Roch, M. (**2026**). "PAMGuard: Application software for passive acoustic detection, classification, and localisation of animal sounds," J. Acoust. Soc. Am. **159**, 437–443.

Hartmann, D., Palmqvist, V., and Stedt, J. (**2025**). "Programmed unmanned aerial vehicles show great potential for monitoring marine megafauna in specific areas of interest," Remote Sens. Ecol. Conserv. **12**, 368–387.

Hermannsen, L., Ladegaard, M., Tønnesen, P., Malinka, C., Beedholm, K., Tougaard, J., Rojano-Doñate, L., Tyack, P., and Madsen, P. (**2025**). "High-frequency vessel noise can mask porpoise echolocation," J. Exp. Biol. **228**, jeb249963.

Ivanchikova, J., and Tregenza, N. (**2023**). "Validation of the F-POD—A fully automated cetacean monitoring system," PLoS One **18**, e0293402.

Jansen, O. E., Aarts, G. M., Das, K., Lepoint, G., Michel, L., and Reijnders, P. J. H. (**2012**). "Feeding ecology of harbour porpoises: Stable isotope analysis of carbon and nitrogen in muscle and bone," Mar. Biol. Res. **8**, 829–841.

Johnson, M. P., and Tyack, P. L. (**2003**). "A digital acoustic recording tag for measuring the response of wild marine mammals to sound," IEEE J. Oceanic Eng. **28**, 3–12.

Johnson, M., Aguilar De Soto, N., and Madsen, P. (**2009**). "Studying the behaviour and sensory ecology of marine mammals using acoustic recording tags: A review," Mar. Ecol. Prog. Ser. **395**, 55–73.

Kastelein, R., Hoek, L., and Jennings, N. (**2018**). "Seasonal Changes in Food Consumption, Respiration Rate, and Body Condition of a Male Harbor Porpoise (*Phocoena phocoena*)," Aquat. Mamm. **44**, 76–91.

Koblitz, J. C., Wahlberg, M., Stilz, P., Madsen, P. T., Beedholm, K., and Schnitzler, H.-U. (**2012**). "Asymmetry and dynamics of a narrow sonar beam in an echolocating harbor porpoise," J. Acoust. Soc. Am. **131**, 2315–2324.

Kyhn, L. A., Tougaard, J., Teilmann, J., Wahlberg, M., Jørgensen, P. B., and Bech, N. I. (**2008**). "Harbour porpoise (*Phocoena phocoena*) static acoustic monitoring: Laboratory detection thresholds of T-PODs are reflected in field sensitivity," J. Mar. Biol. Assoc. United Kingdom **88**, 1085–1091.

Kyhn, L. A., Tougaard, J., Thomas, L., Duve, L. R., Stenback, J., Amundin, M., Desportes, G., and Teilmann, J. (**2012**). "From echolocation clicks to animal density—Acoustic sampling of harbor porpoises with static dataloggers," J. Acoust. Soc. Am. **131**, 550–560.

Ladegaard, M., and Madsen, P. T. (**2019**). "Context-dependent biosonar adjustments during active target approaches in echolocating harbour porpoises," J. Exp. Biol. **222**, jeb206169.

Linnenschmidt, M., Teilmann, J., Akamatsu, T., Dietz, R., and Miller, L. A. (**2013**). "Biosonar, dive, and foraging activity of satellite tracked harbor porpoises (*Phocoena phocoena*)," Mar. Mammal Sci. **29**, E77–E97.

Macaulay, J. D. J., Malinka, C. E., Gillespie, D., and Madsen, P. T. (**2020**). "High resolution three-dimensional beam radiation pattern of harbour porpoise clicks with implications for passive acoustic monitoring," J. Acoust. Soc. Am. **147**, 4175.

Macaulay, J. D. J., Rojano-Doñate, L., Ladegaard, M., Tougaard, J., Teilmann, J., Marques, T. A., Siebert, U., and Madsen, P. T. (**2023**). "Implications of porpoise echolocation and dive behaviour on passive acoustic monitoring," J. Acoust. Soc. Am. **154**, 1982–1995.

Madsen, P. T., Wisniewska, D., and Beedholm, K. (**2010**). "Single source sound production and dynamic beam formation in echolocating harbour porpoises (*Phocoena phocoena*)," J. Exp. Biol. **213**, 3105–3110.

Marques, T. A., Thomas, L., Martin, S. W., Mellinger, D. K., Ward, J. A., Moretti, D. J., Harris, D., and Tyack, P. (**2013**). "Estimating animal population density using passive acoustics," Biol. Rev. **88**, 287–309.

Møhl, B., and Andersen, S. (**1973**). "Echolocation: High-frequency component in the click of the Harbour Porpoise (Phocoena ph. L)," J. Acoust. Soc. Am. **54**, 1368–1372.

Nuuttila, H. K., Brundiers, K., Dähne, M., Koblitz, J. C., Thomas, L., Courtene-Jones, W., Evans, P. G. H., Turner, J. R., Bennell, J. D., and Hiddink, J. G. (**2018**). "Estimating effective detection area of static passive acoustic data loggers from playback experiments with cetacean vocalisations," Meth. Ecol. Evol. **9**, 2362–2371.

Osiecka, A. N., Jones, O., and Wahlberg, M. (**2020**). "The diel pattern in harbour porpoise clicking behaviour is not a response to prey activity," Sci. Rep. **10**, 14876.

R Core Team (**2025**). "R: A language and environment for statistical computing," https://www.r-project.org (Last viewed 24/10/2026).

11 November 2026 21:14:56

Roberts, B. L., and Read, A. J. (**2015**). "Field assessment of C-POD performance in detecting echolocation click trains of bottlenose dolphins (*Tursiops truncatus*)," Mar. Mammal Sci. **31**, 169–190.

Rojano-Doñate, L., McDonald, B. I., Wisniewska, D. M., Johnson, M., Teilmann, J., Wahlberg, M., Højer-Kristensen, J., and Madsen, P. T. (**2018**). "High field metabolic rates of wild harbour porpoises," J. Exp. Biol. **221**, jeb185827.

Rojano-Doñate, L., Teilmann, J., Wisniewska, D. M., Jensen, F. H., Siebert, U., McDonald, B. I., Elmegaard, S. L., Sveegard, S., Dietz, R., Johson, M., and Madsen, P. T. (**2024**). "Low hunting costs in an expensive marine mammal predator," Sci. Adv. **10**, eadj7132.

Rone, B. K., Zerbini, A. N., Falcone, E. A., Keene, E. L., and Schorr, G. S. (**2024**). "Distribution, abundance, and density of harbor porpoises in Hood Canal, Washington," J. Wildl. Manag. **88**, e22543.

Sarnocinska, J., Tougaard, J., Johnson, M., Madsen, P. T., and Wahlberg, M. (**2016**). "Comparing the performance of C-PODs and SoundTrap/PAMGUARD in detecting the acoustic activity of harbor porpoises (*Phocoena phocoena*)," presented at the *Fourth International Conference on the Effects of Noise on Aquatic Life*, Dublin, Ireland, p. 070013.

Schaffeld, T., Bräger, S., Gallus, A., Dähne, M., Kruegel, K., Herrmann, A., Jabbusch, M., Ruf, T., Verfuss, U., Benke, H., and Koblitz, J. (**2016**). "Diel and seasonal patterns in acoustic presence and foraging behaviour of free-ranging harbour porpoises," Mar. Ecol. Prog. Ser. **547**, 257–272.

Scheidat, M., Tougaard, J., Brasseur, S., Carstensen, J., van Polanen Petel, T., Teilmann, J., and Reijnders, P. (**2011**). "Harbour porpoises (*Phocoena phocoena*) and wind farms: A case study in the Dutch North Sea," Environ. Res. Lett. **6**, 025102.

Sørensen, P. M., Wisniewska, D. M., Jensen, F. H., Johnson, M., Teilmann, J., and Madsen, P. T. (**2018**). "Click communication in wild harbour porpoises (*Phocoena phocoena*)," Sci. Rep. **8**, 9702.

Stavenow Jerremalm, J. S., Beesau, J., Carlström, J., Eriksson, P., Jessopp, M., Kavanagh, A., Loisa, O., Michel, M., Owen, K., Rasmussen, M. H., Samaran, F., Todd, N. R. E., and Rogan, E. (**2025**). "Large-scale passive acoustic monitoring data shows seasonal and diel diversity in foraging behaviour of harbour porpoises within their distributional range in the Northeast Atlantic," Ecol Evol **15**, e72566.

Stedt, J., Hamel, H., Torres Ortiz, S., Højer Kristensen, J., and Wahlberg, M. (**2024**). "Harbour porpoises are flexible predators displaying context-dependent foraging behaviours," Ecol. Evol. **14**, e70671.

Stedt, J., Wahlberg, M., Carlström, J., Nilsson, P., Amundin, M., Oskolkov, N., and Carlsson, P. (**2023**). "Micro-scale spatial preference and temporal cyclicity linked to foraging in harbour porpoises," Mar. Ecol. Prog. Ser. **708**, 143–161.

Stevenson, B. C., Borchers, D. L., Altwegg, R., Swift, R. J., Gillespie, D. M., and Measey, G. J. (**2015**). "A general framework for animal density estimation from acoustic detections across a fixed microphone array," Methods Ecol. Evol. **6**, 38–48.

Sveegaard, S., Nabe-Nielsen, J., Staehr, K.-J., Jensen, T., Mouritsen, K., and Teilmann, J. (**2012**). "Spatial interactions between marine predators and their prey: Herring abundance as a driver for the distributions of mackerel and harbour porpoise," Mar. Ecol. Prog. Ser. **468**, 245–253.

Todd, N. R. E., Jessopp, M. J., Rogan, E., and Kavanagh, A. S. (**2025a**). "Multi-year acoustic monitoring reveals fine-scale habitat preferences of harbour porpoise (*Phocoena phocoena*) and species decline within a Special Area of Conservation," Estuarine. Coastal Shelf Sci. **324**, 109470.

Todd, N. R. E., Kavanagh, A. S., Jessopp, M. J., Verboom, W., and Rogan, E. (**2025b**). "Can you hear me? Playback experiment highlights detection range differences between commonly used PAM devices: C-POD, F-POD and SoundTrap," PLoS One **20**, e0320925.

Todd, V. L. G., Pearse, W. D., Tregenza, N. C., Lepper, P. A., and Todd, I. B. (**2009**). "Diel echolocation activity of harbour porpoises (*Phocoena phocoena*) around North Sea offshore gas installations," ICES J. Mar. Sci. **66**, 734–745.

Tollit, D., Joy, R., Wood, J., Redden, A., Booth, C., Boucher, T., Porskamp, P., and Oldreive, M. (**2019**). "Baseline presence of and effects of tidal turbine installation and operations on harbour porpoise in Minas Passage, Bay of Fundy, Canada," J. Ocean Technol. **14**, 24–48.

Torres Ortiz, S., Stedt, J., Midtiby, H. S., Egemose, H. D., and Wahlberg, M. (**2021**). "Group hunting in harbour porpoises (Phocoena phocoena)," Can. J. Zool. **99**, 511–520.

Tregenza, N. (**2013**). "C-POD: Validating cetacean detections," Chelonia Ltd.

Tyack, P. L., Zimmer, W. M. X., Moretti, D., Southall, B. L., Claridge, D. E., Durban, J. W., Clark, C. W., D'Amico, A., DiMarzio, N., Jarvis, S., McCarthy, E., Morrissey, R., Ward, J., and Boyd, I. L. (**2011**). "Beaked Whales Respond to Simulated and Actual Navy Sonar," PLoS One **6**, e17009.

van Beest, F. M., Teilmann, J., Dietz, R., Galatius, A., Mikkelsen, L., Stalder, D., Sveegaard, S., and Nabe-Nielsen, J. (**2018**). "Environmental drivers of harbour porpoise fine-scale movements," Mar. Biol. **165**, 95.

Veneruso, G., Chapuis, L., Hastie, G. D., Vay, L. L., and Cordes, L. S. (**2025**). "Tidal flow masks acoustic detections of harbour porpoises (*Phocoena phocoena*): Implications for passive acoustic studies of cetaceans," J. Acoust. Soc. Am. **158**, 2883–2891.

Voß, J., Rose, A., Kosarev, V., Vílela, R., van Opzeeland, I. C., and Diederichs, A. (**2023**). "Response of harbor porpoises (*Phocoena phocoena*) to different types of acoustic harassment devices and subsequent piling during the construction of offshore wind farms," Front. Mar. Sci. **10**, 1128322.

Warren, V. E., Marques, T. A., Harris, D., Thomas, L., Tyack, P. L., Aguilar de Soto, N., Hickmott, L. S., and Johnson, M. P. (**2017**). "Spatio-temporal variation in click production rates of beaked whales: Implications for passive acoustic density estimation," J. Acoust. Soc. Am. **141**, 1962–1974.

Wisniewska, D. M., Johnson, M., Teilmann, J., Rojano-Doñate, L., Shearer, J., Sveegaard, S., Miller, L. A., Siebert, U., and Madsen, P. T. (**2016**). "Ultra-high foraging rates of harbor porpoises make them vulnerable to anthropogenic disturbance," Curr. Biol. **26**, 1441–1446.

Wisniewska, D. M., Johnson, M., Teilmann, J., Siebert, U., Galatius, A., Dietz, R., and Madsen, P. T. (**2018**). "High rates of vessel noise disrupt foraging in wild harbour porpoises (*Phocoena phocoena*)," Proc. R Soc. B **285**, 20172314.

Wisniewska, D., Ratcliffe, J., Beedholm, K., Christensen, C., Johnson, M., Koblitz, J., Wahlberg, M., and Madsen, P. T. (**2015**). "Range-dependent flexibility in the acoustic field of view of echolocating porpoises (*Phocoena phocoena*)," eLife Sci. **4**, e05651.

Wood, S. N. (**2010**). "Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models," J. R. Stat. Soc. Ser. B. Stat. Methodol. **73**, 3–36.

11 November 2026 21:14:56